Coffee Shop Sales: A Data-Driven Analysis for Business Intelligence

Ayush Dubey
Dept. of CSE/IT
Lovely Professional University
Phagwara, India
Registration No. 12408876

Maneet Kaur (Supervisor)
Dept. of CSE/IT
Lovely Professional University
Phagwara, India

Abstract

This project focuses on the development of a comprehensive Coffee Shop Sales Dashboard designed to convert large volumes of raw transactional data into meaningful and actionable business insights. The study utilizes a dataset comprising 149,116 transaction records from a fictional coffee shop chain operating across multiple locations in New York City. Through the application of data preprocessing techniques, feature engineering, and visualization tools in Microsoft Excel, the project aims to monitor daily and periodic sales performance, identify best-selling and underperforming products, and analyze customer purchasing behavior across different store locations. The dashboard provides a structured and interactive representation of key performance indicators (KPIs), including revenue trends, peak hours, and product demand. The findings offer valuable insights that can assist business managers in making informed decisions related to inventory management, workforce allocation, and marketing strategies. Ultimately, this project demonstrates how business intelligence tools can enhance operational efficiency and profitability in the retail food and beverage sector.

Keywords—Business Intelligence, Data Visualization, Coffee Shop Sales, Excel Dashboard, Retail Analytics

I. Introduction

In today's highly competitive and customer-oriented food and beverage industry, coffee shops must go beyond simply delivering high-quality products to sustain growth and profitability. With the increasing number of competitors and rapidly changing consumer preferences, businesses are required to adopt more strategic and analytical approaches to remain relevant in the market. Customers today expect not only quality beverages but also quick service, personalized experiences, and consistency across visits. As a result, relying solely on traditional business practices or intuition is no longer sufficient. Instead, organizations must leverage data-driven decision-making to gain deeper insights into their operations and customer behavior.

Data-driven decision-making has emerged as a crucial component of modern business intelligence. Coffee shops generate large volumes of transactional data on a daily basis, including information related to sales, product categories, customer preferences, and purchasing patterns. When properly analyzed, this data can reveal valuable insights that help businesses identify trends, predict demand, and optimize their overall performance. The ability to transform raw data into meaningful information allows managers to make informed decisions that can improve efficiency, reduce costs, and enhance customer satisfaction.

This project focuses on designing and developing a dynamic and interactive dashboard that visually represents the sales performance of a coffee shop.

The dashboard serves as a centralized platform where complex datasets are organized and presented in a simplified and user-friendly manner. By systematically structuring transactional data—such as purchase dates, product categories, quantities sold, and pricing details—the dashboard enables stakeholders to gain a comprehensive understanding of day-to-day business operations. Visualization tools such as charts, graphs, and filters further enhance the interpretability of data, making it easier to identify patterns and anomalies.

The primary objective of this project is to convert raw and unstructured data into actionable insights that can support strategic business decisions. The dashboard is designed to provide a multi-dimensional view of the data, allowing users to analyze performance from different perspectives. These include time-based trends, which help in understanding seasonal and daily variations in sales; product performance analysis, which identifies best-selling and low-performing items; and location-based comparisons, which highlight differences in performance across various outlets. Such comprehensive analysis ensures that all critical aspects of the business are taken into account.

The key objectives of the study are as follows:

- To track total sales and revenue across daily, weekly, and monthly timeframes in order to monitor financial performance and growth trends.
- To identify top-performing and underperforming products, enabling better decision-making in terms of inventory management, pricing strategies, and product offerings.
- To analyze customer purchasing behavior and preferences, which can help businesses tailor their products and services to meet customer expectations.
- To determine peak business hours and high-traffic days, thereby assisting in efficient workforce planning and resource allocation.

By achieving these objectives, the project highlights the importance of integrating data analytics into business operations. It demonstrates how the use of dashboards and visualization techniques can significantly enhance operational efficiency by providing clear and actionable insights. Furthermore, it contributes to improving customer satisfaction by enabling businesses to respond effectively to customer needs and preferences. Overall, this study emphasizes that the adoption of data-driven strategies is essential for ensuring long-term sustainability and competitiveness in the modern coffee shop industry.

II. Methodology and Preprocessing

The effectiveness of any data-driven analysis largely depends on the quality, structure, and preparation of the dataset being used. In this project, the dataset has been sourced from Maven Analytics and comprises a total of 149,116 transaction records. These records are distributed across 11 key attributes, including transaction ID, transaction date and time, store location, product category, product type, quantity sold, and unit price. The dataset represents detailed transactional activities of a coffee shop chain, making it suitable for conducting comprehensive sales and customer behavior analysis.

Before performing any analysis or visualization, it is essential to preprocess the data to ensure its accuracy, consistency, and usability. Raw datasets often contain missing values, inconsistencies, and redundant information, which can lead to misleading results if not properly handled. Therefore, a systematic methodology was followed to clean, transform, and enhance the dataset for meaningful analysis.

A. Data Pre-processing

Data preprocessing is a crucial step in transforming raw data into a structured format suitable for analysis. It ensures that the dataset is free from errors, inconsistencies, and irrelevant information.

Data Cleaning:
The initial step involved identifying and correcting inconsistencies within the dataset. Duplicate entries, irrelevant columns, and formatting errors were carefully examined and removed to streamline the data. This process helped in reducing noise and improving the overall quality of the dataset. Additionally, inconsistencies in naming conventions and data formats were standardized to maintain uniformity across all records.

Handling Missing Values:
Missing or null values are common in large datasets and can significantly impact the results of analysis. In this project, missing entries in fields such as product size and product category were handled using appropriate techniques. Where possible, logical imputation was applied based on available information. In cases where imputation was not feasible, the values were categorized as “Not Defined.” This approach ensured that no data was discarded unnecessarily while maintaining analytical integrity.

B. Feature Engineering

Feature engineering plays a vital role in enhancing the analytical capabilities of a dataset by creating new variables that provide deeper insights into the data.

Time-Based Feature Extraction: Since the dataset includes timestamp information, several time-related features were derived to facilitate temporal analysis. These include extracting the month name, day of the week, and hour of the day from the transaction timestamps. This transformation allows for the identification of seasonal trends, weekly patterns, and peak business hours, which are essential for operational planning and decision-making.

Sales Metric Creation: To evaluate financial performance, a new variable named *Total_bill* was created. This metric represents the total revenue generated from each transaction and was calculated using the formula:

Total_bill = Quantity × Unit Price

This derived feature serves as a key performance indicator (KPI) for analyzing revenue trends and comparing sales across different dimensions.

Customer Behavior Metrics: To better understand customer purchasing patterns, additional metrics were calculated. The *Average Bill per Person* was determined to be $4.69, indicating the typical amount spent per transaction. Similarly, the *Average Order per Person* was calculated as 1.44 items, reflecting the average number of products purchased in a single transaction. These metrics provide valuable insights into customer spending behavior and help in designing pricing and promotional strategies.

The implementation of these engineered features significantly enhanced the dataset's ability to support detailed and multi-dimensional analysis.

C. Tools and Techniques

The analysis and visualization of the dataset were conducted using Microsoft Excel, which provides a wide range of tools for data manipulation and business intelligence.

Data Analysis Tools: Pivot Tables were extensively used to summarize and aggregate large volumes of data efficiently. They enabled the grouping of data based on various dimensions such as product category, time, and location, thereby simplifying complex datasets into meaningful summaries.

Data Visualization Techniques: Pivot Charts were utilized to visually represent trends and patterns in the data. These charts made it easier to interpret results and communicate findings effectively. Slicers were incorporated to add interactivity to the dashboard, allowing users to filter data dynamically based on specific criteria such as date, product type, or store location.

Dashboard Design: Conditional formatting techniques were applied to highlight key insights, such as top-performing products or peak sales periods. The combination of these tools resulted in the creation of an interactive and user-friendly dashboard that supports real-time data exploration and decision-making.

Overall, the methodology adopted in this project ensures that the dataset is clean, structured, and enriched with meaningful features. This robust preprocessing and analytical framework forms the foundation for accurate insights and effective business intelligence outcomes.

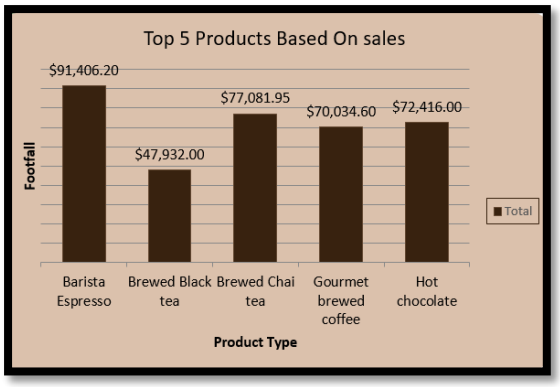
III. Results and Analysis

The analysis of the coffee shop sales dataset provides valuable insights into business performance, customer behavior, and operational efficiency. By leveraging the processed data and dashboard visualizations, multiple dimensions of analysis were explored, including product performance, temporal trends, and location-based variations.

A. Sales Performance by Category and Product

The analysis reveals that coffee-based products are the dominant contributors to overall revenue, accounting for approximately 38% of total sales. This indicates that the core strength of the business lies in its primary offerings, particularly traditional and specialty coffee beverages.

Among individual products, *Barista Espresso* emerged as the top-performing item, generating a total revenue of $91,406.20. This highlights its strong demand and popularity among customers. Other notable contributors include *Brewed Chai Tea* and *Hot Chocolate*, which also demonstrate consistent sales performance and customer preference.



The concentration of revenue among a few key products suggests that these items play a critical role in sustaining business profitability. At the same time, it indicates the need to carefully monitor underperforming products, which may require strategic interventions such as promotional offers, repositioning, or potential removal from the menu.

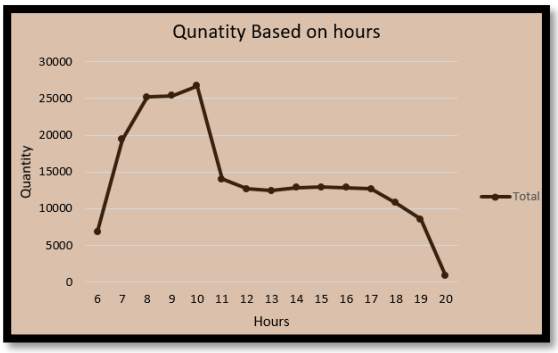
Overall, the findings emphasize the importance of maintaining product quality, ensuring consistent availability, and focusing on high-demand items to maximize revenue generation.

B. Temporal Trends

The temporal analysis of sales data provides insights into how customer behavior varies across different times of the day and days of the week.

The data indicates that peak sales occur between 7:00 AM and 9:00 AM, which aligns with typical morning routines, including work commutes and breakfast consumption. This period represents the highest customer demand, making it a critical time for business operations. Ensuring adequate staffing, efficient service, and product availability during these hours is essential for maximizing sales and maintaining customer satisfaction.

Weekly trends further reveal that Tuesday and Wednesday record the highest sales volumes, indicating strong mid-week demand. In contrast, Sunday experiences the lowest customer footfall, suggesting reduced activity during weekends.



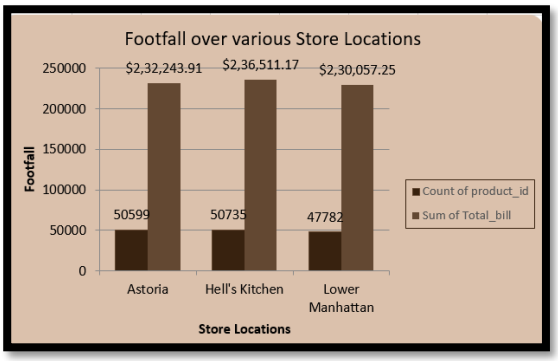
These patterns provide actionable insights for business optimization. For instance, promotional strategies can be implemented during off-peak hours or low-demand days to stimulate sales. Similarly, staffing schedules can be adjusted to align with peak demand periods, thereby improving operational efficiency and reducing unnecessary labor costs.

C. Location-Based Performance

The analysis of sales performance across different store locations highlights significant variations in customer engagement and revenue generation.

Among the analyzed locations, Hell's Kitchen stands out as the top-performing outlet in terms of both total sales and customer footfall. This suggests that the location benefits from favorable factors such as high population density, better accessibility, or stronger brand presence.

On the other hand, Astoria records comparatively lower sales performance, indicating potential challenges such as reduced customer traffic, less effective marketing, or unfavorable location dynamics.



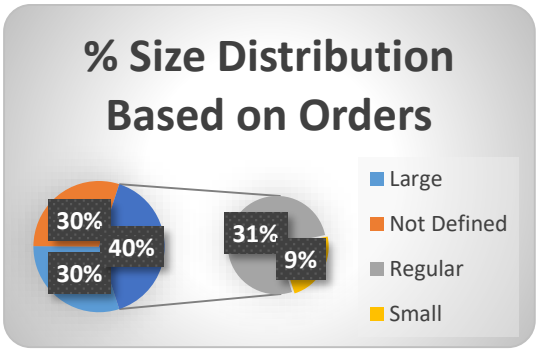
These insights are crucial for developing targeted, location-specific strategies. High-performing locations can be further leveraged through expansion or premium offerings, while

underperforming outlets may require promotional campaigns, pricing adjustments, or operational improvements to boost performance.

D. Size Distribution Analysis

The distribution of order sizes was analyzed using a composite pie-of-pie chart to provide granular insight into product dimensions. As illustrated in Fig. 1, the primary categories of Large and Not Defined each account for 30% of the total order volume. The remaining 40% of the distribution is further decomposed into a secondary sub-chart to distinguish between Regular and Small sizes.

Within this segment, Regular sizes constitute the majority at 31%, while Small sizes represent a minority share of 9%. This breakdown suggests a significant consumer preference for larger or standard-sized products, while also highlighting a notable portion of data (30%) where size parameters remain unspecified, indicating a potential area for improved data logging.



E. Temporal Order and Sales Trends

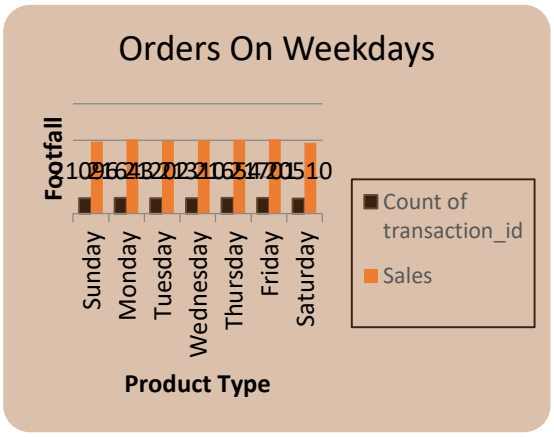
To evaluate operational performance across a weekly cycle, transaction counts and sales volumes were plotted against weekdays (Fig. 2). The data reveals a remarkably consistent footfall and sales performance throughout the week, with marginal fluctuations.

Peak Activity: The highest volume of transactions was recorded on Friday (21,701), closely followed by Thursday (21,654) and Monday (21,643).

Minimum Activity: A slight decline in performance is observed on Saturday, which recorded the lowest transaction count of 20,510.

Sales Correlation: The "Sales" metric (represented by the orange indicators) demonstrates a strong positive correlation with the "Count of transaction_id." The uniformity of the bars across all seven days suggests a stable demand environment, devoid of significant weekend or mid-week surges.

F. Revenue Contribution by Product Category

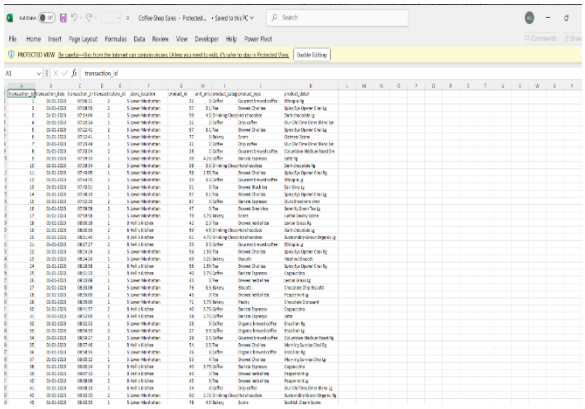


A comprehensive analysis of sales revenue across various product categories was performed to identify primary growth drivers. As depicted in Fig. 3, the sales distribution is heavily dominated by beverage-related products, specifically within the coffee and tea segments.

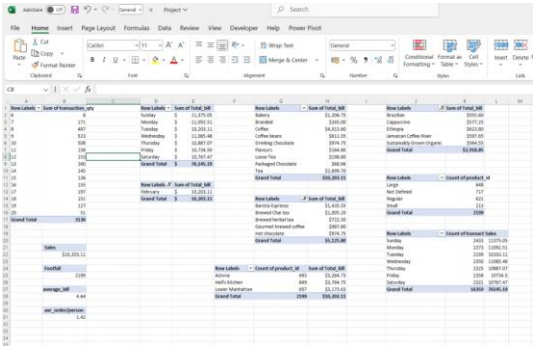
- Market Dominance: Coffee serves as the primary revenue generator, accounting for 39% of total sales. When combined with Tea (28%) and Coffee beans (6%), the total beverage category represents a commanding 73% of the overall sales volume.
- Secondary Categories: The Bakery segment holds a significant market share of 12%, indicating a strong consumer preference for food pairings with beverages. Drinking Chocolate also contributes a notable 10% to the revenue stream.
- Niche Segments: The remaining categories, including Branded (2%), Flavours (1%), Loose Tea (1%), and Packaged Chocolate (1%), represent niche segments with minimal contribution to the current sales mix.

The data suggests that the business model is highly specialized in the beverage industry, with coffee and tea acting as the foundational pillars of the commercial operation. The relatively high performance of the Bakery section suggests opportunities for cross-promotion and bundled offerings to further leverage the high footfall associated with beverage sales.

Raw Data



Pivot Table



Overall Findings

The operational analysis reveals a high-performing enterprise with total sales of **$698,812.33** and a total footfall of **149,116**. Performance is characterized by a significant morning surge, with transaction quantities peaking between **08:00 and 10:00** before stabilizing into a consistent midday plateau. This demand is met with remarkable uniformity across all store locations; **Hell's Kitchen** serves as the primary revenue driver at **$236,511.17**, though **Astoria** and **Lower Manhattan** follow closely, indicating a balanced and successful geographical scaling strategy.

The financial backbone of the business is firmly rooted in its beverage offerings, with **Barista Espresso** leading individual product sales at **$91,406.20**. When synthesized with the category data, it is evident that coffee and tea products constitute the vast majority of revenue, while the **12%** bakery contribution represents a secondary but stable growth lever. Furthermore, the stability of daily transaction counts—peaking at **21,701** on Fridays—suggests a mature business model supported by a loyal, recurring customer base rather than erratic market spikes. Collectively, these metrics point toward a highly optimized operation with predictable demand cycles and consistent cross-location performance.

Linkedin Link

www.linkedin.com/in/engineer-ayush

IV. Discussion and Future Scope

The findings from the analysis provide meaningful insights into both the strengths and limitations of the current system. One of the most significant observations is that approximately 31% of transactions were categorized as "Not Defined" in terms of product size. This indicates a substantial gap in data entry practices, which can negatively impact the accuracy and reliability of analysis.

From a data management perspective, incomplete or inconsistent data reduces the effectiveness of business intelligence systems. Therefore, it is essential to implement standardized data entry protocols and validation mechanisms to ensure high-quality data collection. Improving data accuracy will lead to more precise analysis and better decision-making outcomes.

Future Scope

The scope of this project can be significantly expanded by incorporating advanced technologies and analytical techniques:

- Integration with Real-Time POS Systems:
  Connecting the dashboard with real-time Point of Sale systems would enable continuous data updates, allowing businesses to monitor performance instantly and respond quickly to changing conditions.

- Predictive Analytics: The use of predictive models can help forecast future demand based on historical trends, seasonal variations, and external factors such as weather conditions or holidays. This would support proactive decision-making and inventory planning.
- Advanced Analytical Tools: Implementing tools such as Power BI, Python, or machine learning algorithms can provide deeper insights and enable more sophisticated analysis, including customer segmentation and demand forecasting.
- Enhanced Data Quality Measures: Introducing automated validation systems and structured data input methods can significantly reduce errors and improve the reliability of future analyses.

Conclusion

The Coffee Shop Sales Dashboard developed in this project serves as an effective business intelligence solution for transforming raw transactional data into actionable insights. By analyzing various aspects such as sales trends, product performance, customer behavior, and location-based variations, the dashboard provides a comprehensive overview of business operations. The study demonstrates that data-driven decision-making plays a critical role in improving operational efficiency, optimizing resource allocation, and enhancing overall profitability. The insights derived from the analysis enable businesses to make informed strategic decisions, such as identifying high-performing products, optimizing staffing schedules, and implementing targeted marketing strategies.

Furthermore, the project highlights the importance of data quality and the need for continuous improvement in data collection and analysis processes. With the integration of advanced technologies and real-time analytics, such systems can become even more powerful and impactful.

In conclusion, the adoption of data visualization and business intelligence tools is essential for ensuring long-term sustainability, competitiveness, and growth in the modern coffee shop industry.

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